

1. Document Overview & Executive Summary

This comprehensive documentation reference covers the complete architecture, file tree, components, services, database models, system requirements, and security frameworks for MeDo ERP v4.5 Enterprise. It serves as an official technical manual for developers, systems managers, and corporate auditors.

2. System Architecture & File Tree Overview

- Root Configuration: index.html, package.json, server.ts, vite.config.ts, tsconfig.json, firestore.rules
- src/components: Over 100 modular enterprise views including App.tsx, Dashboard.tsx, ChartOfAccountsView.tsx, JournalEntriesView.tsx, SalesAndReturnsView.tsx, PurchasesAndReturnsView.tsx, InventoryView.tsx, MohasibDataMigrationWizard.tsx
- src/services & utils: Gemini AI Advisor engine, voice-search parser, email dispatchers, ZATCA Phase 2 QR helpers
- src/db & docs: PostgreSQL Drizzle schemas, multi-currency ledger engines, legal compliance and audit manuals

3. Technical Stack & Dependencies

- Frontend Stack: React 19, Vite 6, TypeScript 5.8, Tailwind CSS v4, Motion (Framer Motion)
- Backend & Storage: Express v4 Server, tsx Runtime Engine, PostgreSQL Cloud SQL + Drizzle ORM, Firebase Firestore
- AI Integration: @google/genai SDK powered by Gemini 3.7 Flash & 3.6 Flash models
- Security & Biometrics: WebAuthn Passkeys, Sovereign WAF Shield, Audit Trail Tracker
- Document Engines: HTML2PDF.js, PDFKit, XLSX Excel Export, DOCX Builder

4. System & Server Requirements

- Server Environment: 2+ vCPU, 4GB+ RAM, Node.js 20+, Port 3000 with SSL/TLS reverse proxy
- Database: PostgreSQL 15+ instance with multi-tenant state isolation and automated backups
- Multi-Currency Engine: Native support for YER (Sanaa: 530 / Aden: 1920), SAR (3.75), and USD

5. Governance, Audit & Legal Compliance

- Immutable Audit Trail: Every transaction logged with cryptographically unique audit IDs and IP signatures.
- IFRS & ZATCA Compliance: Fully compliant with IFRS accounting standards and ZATCA Phase 2 tax requirements.
- Legal Disclaimer: MeDo ERP is engineered for enterprise financial management; final audit signoffs remain under client CFO authority.